

APE LEVEL

GAMING & BRANDING

RAGE REALMZ / WHISPER MARKET

WHISPER RECONCILES ITS FIRST RESOURCE TRANSACTION

Polygon Amoy testnet | Technical development report

October 30, 2026

PAYOUT	RESOURCE	TESTS
1 tTILT	1 WATER	45 / 45

Verified blockchain payout • Verified receipt • Finalized resource transfer

Executive overview

Ape Level Gaming & Branding has completed and reconciled its first verified Whisper Market resource transaction on Polygon Amoy. A RAGE REALMZ player sold one individually identified unit of Water, received exactly one tTILT from the deployed Whisper V2 settlement contract, and had that Water transferred into the market treasury's authoritative resource ledger after independent on-chain receipt verification.

The achievement extends beyond a wallet transfer. It joins multiplayer wallet authentication, confirmed inventory, resource reservations, canonical trade manifests, server-authorized settlement signatures, smart-contract payments, receipt verification and idempotent off-chain ownership changes into a single demonstrable transaction. The initial browser confirmation failed to complete, exposing a recoverability gap that was successfully resolved without a second payment.

MEASURE	VERIFIED RESULT
Blockchain outcome	SUCCESS • Block 51,535,015
Financial payout	1 tTILT to player wallet
Resource transfer	1 Water transferred to Whisper treasury
Final settlement status	FINALIZED
Regression tests	45 passed / 0 failed
Source control	Commit 9f5269b, pushed to feature/polygon-wallet-auth

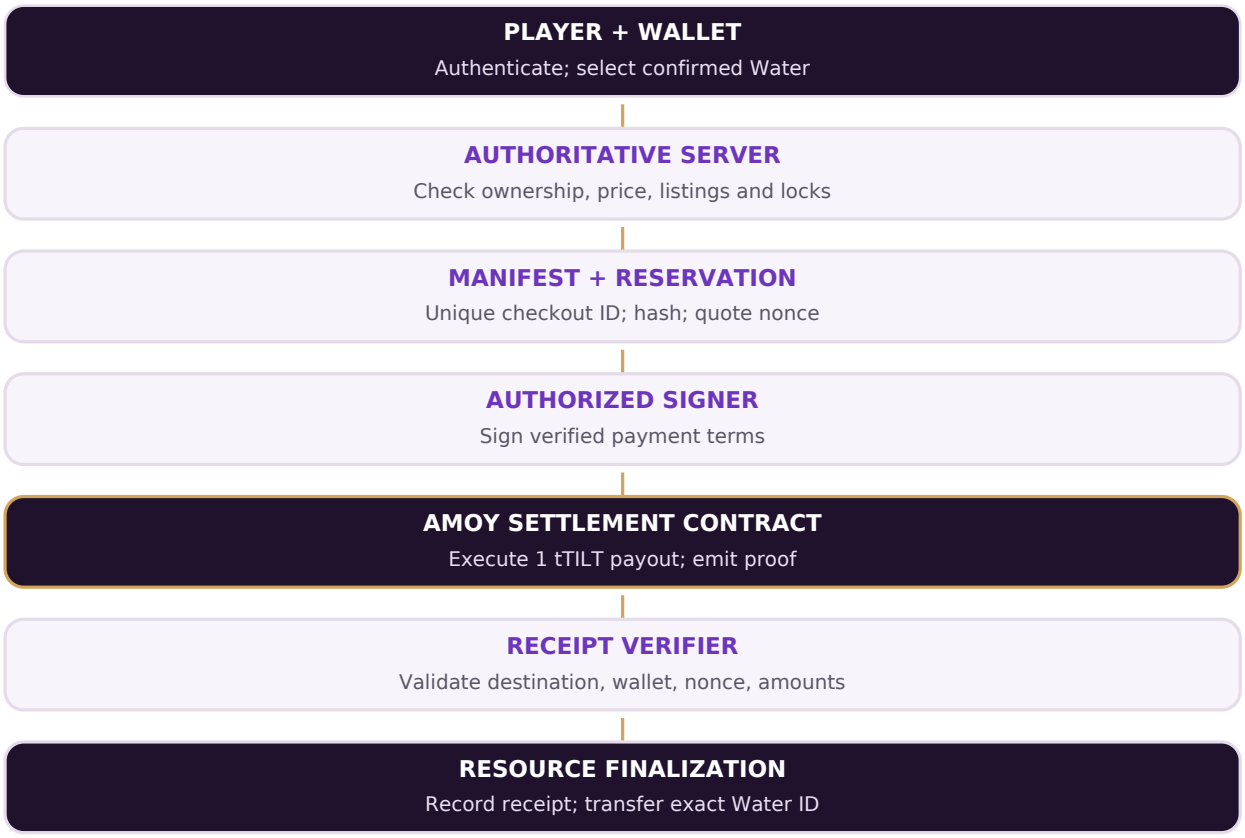
1 | Why Whisper was built this way

Whisper is intended to support a persistent resource-based economy without minting every Water, Wood, Ore or Wheat unit as a wallet-held NFT. Each resource remains an individually identifiable object in the game's authoritative ledger, while tTILT is an ERC-20-style settlement currency on Polygon Amoy during testing. This avoids an on-chain transaction for every inventory action, while preserving a public financial proof for trades.

The architecture deliberately separates payment truth (what the blockchain executed) from resource truth (which exact unit the game owns or transfers). A safe market cannot infer one from the other. A receipt must be matched to the specific authorized manifest, and inventory must not move until receipt verification succeeds.

2 | End-to-end architecture

The operational path below shows the successful sale and its reconciliation. The automated recovery stage is still being improved; the first sale required a verified manual recovery using existing backend functions.



This design does not put resource IDs or inventory balances into a wallet-held ERC-1155 Water token. Settlement on Amoy and resource ownership in Whisper are linked by a manifest hash, a signed quote, the transaction receipt and the internal transfer record.

3 | Creating a verifiable resource sale

Authenticated inventory selection

The player connected wallet 0x9A63d079c807Ead4e241aC2639c46f62F2Aa0397 and requested a one-Water sale. The server selected confirmed, AVAILABLE Water owned by that wallet. Selection excludes active listings, cart reservations and conflicting settlement locks; ownership and availability are checked again inside the reservation transaction.

The exact Water unit ultimately finalized was water-2338a9c9-b59e-4ec1-9d41-23e5736fe5a1. That resource ID, not merely a quantity of one, is what the ledger records as moving from player to treasury.

Canonical manifest and economic terms

MANIFEST FIELD	VALUE
Direction / resource	SELL / water
Resource quantity	1 uniquely identified unit
Unit price	1,000,000,000,000,000,000 Dust
Payment to treasury	0 Dust
Payment to player	1,000,000,000,000,000,000 Dust (1 tTILT)
Chain	Polygon Amoy (80002)

Whisper canonicalizes line items and resource IDs before computing a Keccak-256 hash. The on-chain checkout binds signed amounts, nonce, deadline, player and manifest hash; the server can later reconstruct the same manifest to detect tampering. The resulting settlement hash was 0xbd18e16216cc3436292974ec10b65063e42ef1ad1d1082d1c0a74caf3af2b07b.

4 | Solving the duplicate manifest collision

The first failure was a database UNIQUE constraint on whisper_settlement_intents.manifest_hash. With the original deterministic V1 format, selling the same Water at the same price generated the same hash after a previous checkout expired or was cancelled. Settlement identifiers changed, but they were absent from the hashed economic manifest. Historical records therefore caused legitimate retries to collide.

Ape Level introduced V2 manifests containing a server-generated unique checkout identifier. Reservation creation and every verification path were updated together, while V1 reconstruction was retained for historical records. The database uniqueness constraint and transaction integrity checks were not removed. This was tested across the manifest builder, signer, sale finalizer and buy finalizer.

5 | The first live blockchain execution

The authorized checkout was submitted directly to the deployed Whisper V2 contract. The Amoy transaction succeeded in block 51,535,015, using 89,867 gas units. The tTILT token emitted a transfer of exactly 1 tTILT from the Whisper settlement contract to the player wallet. The transaction's caller, destination, manifest hash, nonce and payout matched the stored reservation.

ON-CHAIN EVIDENCE	RESULT
Transaction	0xe85fce89b2f388f8f6a908394745e68b7faa81122ef67b084e7a4b6d8ab55aed
Settlement contract	0xb2b6996D5fd04E24c8065df70D801EbAa7F50C29
Token contract	0xc1dD560375459B19D6437F076F1708E286Bb5D67
Sender / recipient	0x9A63d079c807Ead4e241aC2639c46f62F2Aa0397
Payment direction	Contract → player
Payout / gas	1 tTILT / 89,867 gas units

Public transaction evidence: View the Amoy transaction on Polygonscan. This proof establishes blockchain execution and payment, not by itself the completion of the separate resource ledger.

6 | The reconciliation challenge

After the payout, the database still showed intent cd0e158e-6045-422f-a2b2-47c9006a2c66 as RESERVED with no transaction hash, receipt or Water transfer. The chain had paid; the game had not recorded delivery. This exposed a real integration failure: browser-side confirmation can time out or disconnect after execution, while an automatic recovery worker limited to PENDING checkouts cannot discover a still-RESERVED but already-paid transaction.

The team investigated the WebSocket confirmation handler, pending transition, two-confirmation receipt verifier, durable receipt ledger, recovery worker and settlement finalizer. Rather than overwrite inventory or pay again, it reprocessed the original transaction using Whisper's existing verification and finalization modules.

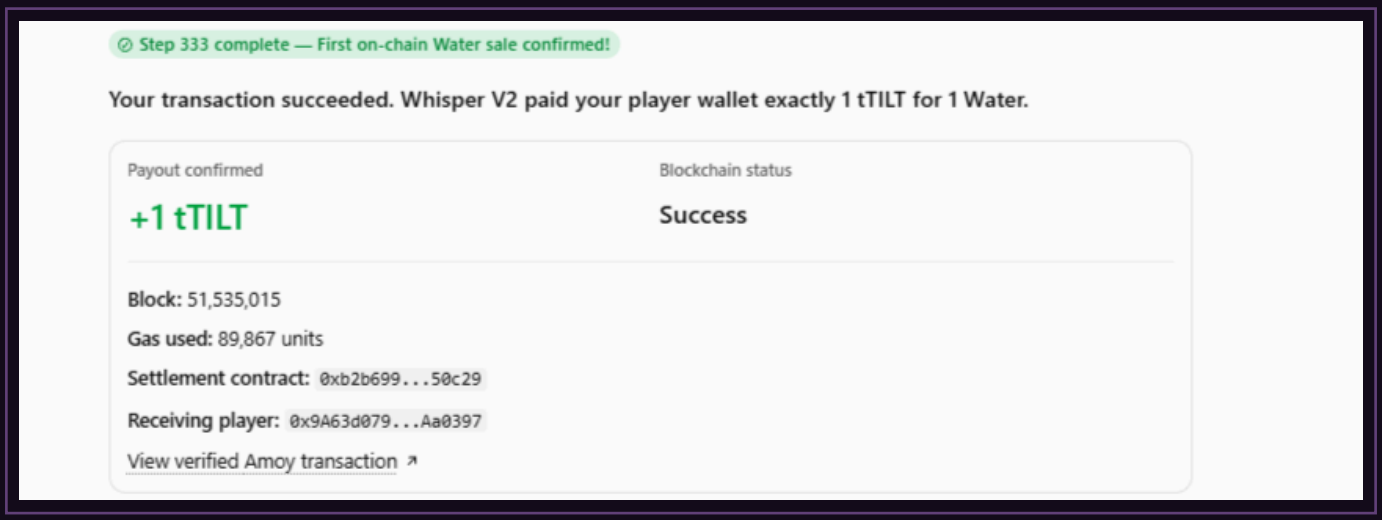
What the verified repair did

- Matched the original settlement intent to the transaction's manifest, player, nonce, destination and payment amounts.
- Verified the successful Amoy receipt with the existing verification function and at least two required confirmations.
- Linked that already-executed transaction hash to the original reservation and advanced it to PENDING.
- Recorded the verified receipt, including matching settlement proof fields.

FIGURE 1 / FIRST ON-CHAIN EXECUTION

The first tTILT payout is confirmed

The development confirmation records the initial successful Polygon Amoy checkout: one Water sold for 1 tTILT, block 51,535,015 and 89,867 gas units.



Screenshot evidence: the successful blockchain payment was established before the game-side resource ledger had finalized. The transaction proof is the Amoy receipt; this screenshot records the development verification.

• Invoked the original Water sale finalizer, transferring the exact locked resource ID once and marking the intent FINALIZED.

No second contract transaction was executed, and no additional gas payment was required for reconciliation. The finalizer returned finalized: true and transferred: 1; subsequent database inspection confirmed the transfer record and original transaction hash.

7 | Inventory accounting after finalization

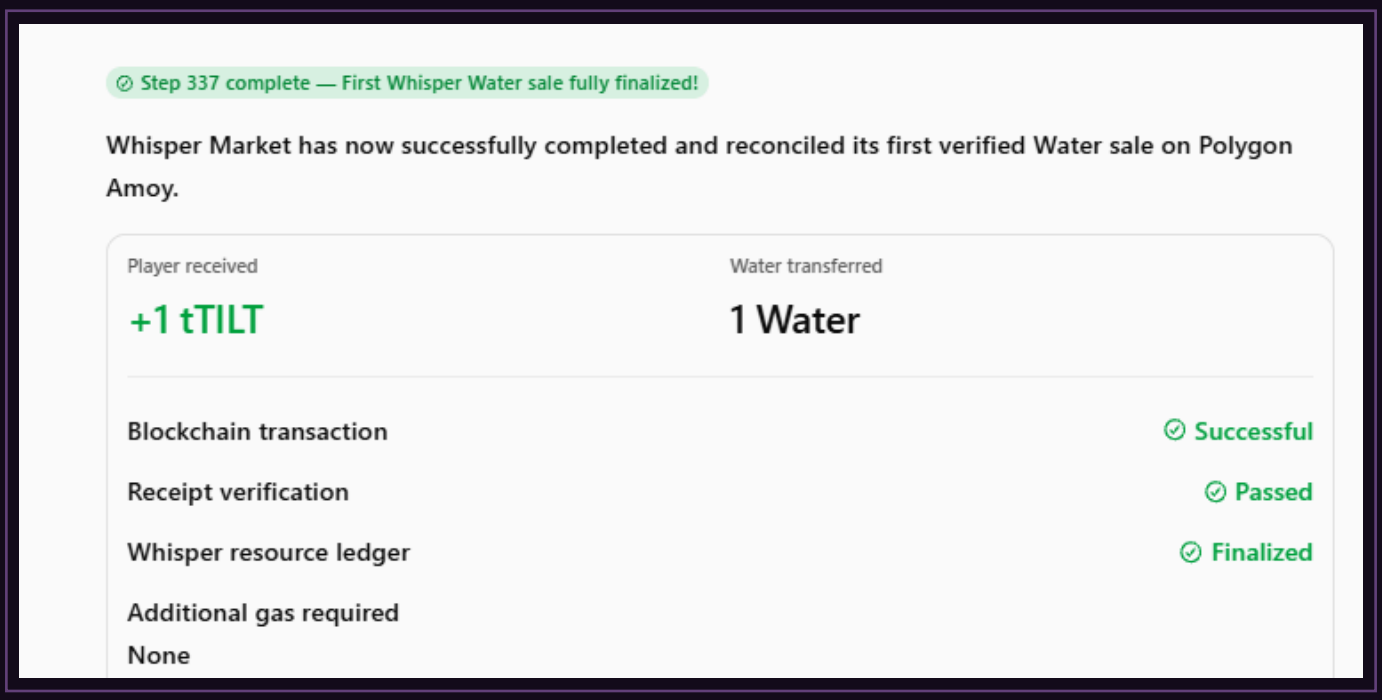
RESOURCE HOLDER	BEFORE	AFTER
Selling player, AVAILABLE Water	12	11
Whisper treasury, AVAILABLE Water	0	1

The transfer record names the exact previous owner, new owner, resource ID and transaction. This is the decisive evidence that the game-side exchange completed, rather than merely displaying a paid token balance.

FIGURE 3 / RECONCILIATION COMPLETION

One transaction. One finalized resource.

Following independent receipt verification and ledger recovery, the game recorded exactly one Water transfer from the selling player into Whisper?s resource treasury.



Screenshot evidence: Step 337 reports blockchain success, receipt verification passed, and resource-ledger finalization complete. No additional gas was required to reconcile the original payment.

8 | Separating the treasury from its controller

During testing, the market dashboard attributed the controller wallet's personal Water to the treasury. The cause was configuration: the resource treasury lookup pointed to a wallet that also held player assets. A separate inspection showed 22 AVAILABLE Water owned by the controller wallet, zero owned by the settlement contract and zero previously finalized treasury transfers.

The server was changed to use the designated Whisper V2 settlement contract address as the ledger identity for treasury-owned Water. This is a resource-accounting address, not a claim that Water itself is tokenized and stored in that contract. Controller permissions and the controller's personally owned resources are distinct from the market's acquired inventory.

ROLE	ACCOUNTING RULE
Controller wallet	Personal Water/tTILT remain personal; administrative control alone transfers nothing.
Whisper V2 contract	Holds settlement tTILT; its address identifies treasury Water in the resource ledger.
Verified resource transfer	Moves only the exact Water acquired under a finalized settlement.

The resulting live sale transferred Water from the selling player 0x9A63…0397, not the controller wallet. The UI's treasury liquidity reader already queried token.balanceOf(settlement), so the distinction for contract-held tTILT was correct. The Water resource ownership calculation was the part requiring correction.

9 | Browser confirmation and recovery improvements

The original checkout client used a 20-second response timeout, while the backend required a real Amoy receipt and multiple confirmations. The client was updated to wait up to 120 seconds for confirmation. After receiving a transaction hash, it now retains the active checkout and prefers confirming that existing transaction before preparing another sale.

A persistent local browser record stores the checkout ID and submitted transaction hash across a refresh, and recovery checks that the connected wallet matches the original wallet. The browser supplies an identifier to the server; it does not itself authorize finalization. The server must still verify chain, contract, manifest, nonce, recipient and payment amounts.

Scope and limitation: These browser improvements were syntax-checked and committed, but no second complete live sale was shown testing them. Recovery also remains incomplete for a transaction whose hash is lost before the browser stores it. Extending server-side event discovery and bounded reconciliation of on-chain checkout logs is the next development objective.

10 | Tests and engineering controls

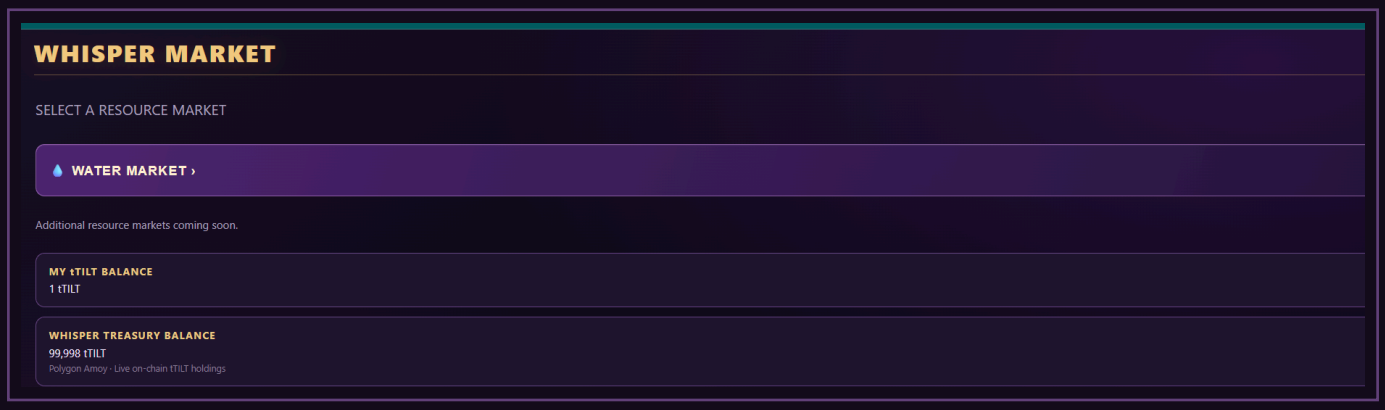
The settlement implementation passed 45 automated tests with zero failures at the commit stage. Test coverage included manifest ordering, duplicate resource rejection, invalid wallets, price changes, exact resource locks, concurrent reservations, expired drafts, pending settlement protection, correct-network enforcement, reverted or missing receipts, wrong senders and destinations, nonce mismatch, amount mismatch, duplicate settlement events, tamper detection and idempotent finalization.

These automated checks are meaningful protections for the tested pathways. They do not substitute for comprehensive mainnet auditing, adversarial review or production operational testing. The testnet-first approach was used specifically to discover and repair integration failures before moving economic activity to production.

FIGURE 2 / VERIFIED MARKET BALANCES

Whisper Market after settlement

The RAGE REALMZ marketplace displays the player?s 1 tTILT balance and Whisper?s live contract-held treasury balance, separately from the off-chain Water ownership ledger.



Screenshot evidence: the interface shows 1 tTILT for the connected player and 99,998 tTILT in Whisper?s on-chain treasury at the captured moment. These token balances are not a substitute for resource ownership records.

11 | Evidence and development provenance

EVIDENCE	DETAIL
Settlement transaction	0xe85fce89...ab55aed • Amoy block 51,535,015
Final settlement intent	cd0e158e-6045-422f-a2b2-47c9006a2c66
Ledger result	FINALIZED; exactly one resource transfer
Source commit	9f5269b10aa3ec8a52c9955f02e216423b749aa9
Remote branch	feature/polygon-wallet-auth
Commit tests	45 passed, 0 failed

Source: View the settlement-development GitHub commit. The push reported nine changed files, 168 insertions and 85 deletions. Unrelated Homestead, release, contract and art files remained uncommitted and were deliberately excluded.

The transaction proof is publicly inspectable on-chain. The corresponding Water ownership record was validated in the private game database; that resource proof is not automatically a public-chain commitment. Future WhisperLog publication and Homestead resource-state commitments are distinct planned proof mechanisms and should not be represented as already completed by this trade.

12 | Next technical priorities

• Implement bounded server-side discovery of Amoy checkout events for successful transactions whose hashes never reach browser storage.

• Exercise a second sale under normal browser confirmation, including a delayed response and reconnect scenario.

• Test treasury-to-player Water purchases at the 1.25 tTILT test price with exact resource ownership transfer and receipt checks.

• Extend the settlement architecture to future resource types without replacing resource-ID verification or treasury separation.

• Complete pre-mainnet hardening, audit and explicit go-live authorization before handling production tTILT funds.

Conclusion

The first reconciled Whisper Market sale demonstrated the required connection between the game economy and blockchain payment: one authenticated seller, one unique Water unit, one authorized contract checkout, one successful 1 tTILT payout, one verified receipt and one finalized ownership transfer.

Equally important, the test discovered concrete failures in manifest uniqueness, treasury identity and post-payment confirmation - and those failures were diagnosed without discarding historical records or executing a duplicate payout. Ape Level now has a demonstrated settlement foundation, clear evidence of what worked, and a defined engineering path toward reliable automated recovery and broader testnet trading.

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